MithunKumar J

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Education

Vellore Institute of Technology, Chennai Chennai, India
Bachelor of Technology in Electronics and Communication Engineering *June 2024 – Present (2028)*

- Relevant Courses: VLSI Design, Microprocessors & Microcontrollers, Analog Communication Systems, Random Processes, Digital Signal Processing, Complex Variables & Linear Algebra

New Prince Shri Bhavani, Chennai Chennai, India
All India Senior School Certificate Examination (Bio-Math Stream) *June 2022 – May 2024*

Apex Pon Vidhyashram, Chennai Chennai, India
All India Secondary School Certificate Examination *June 2012 – May 2022*

Technical Skills

Languages: C, C++, Python, Embedded C, Verilog HDL, 8051/8086 Assembly, ARM Assembly

Embedded & Hardware: ESP32, ESP32-C3, Raspberry Pi, ARM Cortex-M, UART, SPI, I^2C, ESP-NOW, KiCad PCB Design, LTspice, Cadence Virtuoso (GPDK 180nm)

AI / ML / CV: YOLOv8, ByteTrack, MediaPipe, TensorFlow Lite, Random Forest, XGBoost, Resemblyzer, Phi-3 Mini, Ollama, ROS2, Nav2, EKF SLAM

Web / Frontend: Next.js 14, React, GSAP, ScrollTrigger, Three.js, Chart.js, Tailwind CSS

Backend / Automation: Flask, Firebase, Supabase, Google Sheets API, n8n, OpenRouter, MQTT

Tools: Git, Linux, Docker, Figma, Postman, Cadence Virtuoso

Experience

Akvosphere Chennai, India (Hybrid)
Embedded Engineer *2024 – Present*

- Designed production-grade multi-sheet KiCad PCB schematics for an ESP32-C3-WROOM-02 parent board integrating CP2102N USB-UART, MCP73871 battery charger, LM1117 LDO, W25Q32 flash, BME280 sensor, and SSD1306 OLED.
- Engineered I^2C-over-USB-C child board communication: repurposed D+/D- lines for SDA/SCL with USBLC6-2SC6 ESD protection and powered via VBUS rail.
- Architected dual-MCU failover system using TCA9543A I^2C mux and heartbeat GPIO for RPi + ESP32 high-availability control of the atmospheric water generator.
- Identified and resolved critical firmware-hardware conflicts: IO15/IO17 QSPI flash conflicts, floating BME280 SDO address pin, and strapping-pin CS_SD clash.
- Developed ML water-yield prediction system (RandomForest + Flask REST API) and TANGEDCO time-of-use energy optimizer; built NLP query interface via Claude API.
- Implemented n8n automation workflows for AWG telemetry and alert pipelines; integrated OpenRouter multi-LLM routing for smart plant monitoring.

Shaurya Racing (Formula EV Team) VIT Chennai
Electronics Team Member *2024 – Present*

- Contribute to PCB design, schematic review, and embedded systems integration for a Formula Student-style electric vehicle.
- Work on sensor interfacing (current, temperature, hall-effect) and power electronics for EV drivetrain and BMS subsystems.
- Collaborate on vehicle electronics architecture including fault detection logic, driver interface electronics, and real-time telemetry.

Projects

Smart Attendance System | *ESP32, RFID, Face Recognition, Python, Google Sheets API*

- Built dual-factor attendance system fusing RFID card authentication with real-time face verification for tamper-resistant identity confirmation.
- Automated Google Sheets attendance logging via API; triggered email alerts on unauthorized access attempts with timestamped snapshots.

Smart Classroom IoT Monitoring | *ESP32, MAX30100, MPU6050, MLX90614, Firebase, React, Raspberry Pi*

- Deployed multi-sensor node network monitoring student biometrics (heart rate, SpO_2, body temperature) and classroom motion in real-time.
- Streamed sensor data to Firebase Realtime Database; built a React dashboard with live charts, threshold-based alerts, and historical trend views.

UIDAI Aadhaar Analytics Pipeline | *XGBoost, Flask, Python*

- Built anomaly detection pipeline using XGBoost Poisson regression over UIDAI enrollment data; detected geographic and temporal outliers in registration patterns.
- Served insights via Flask analytics dashboard with interactive visualization layer; developed as a team of 3 (Mithun, Tharun, Sarveshvar).

MATLAB Chaos-Based Secure Communication | *MATLAB, DSP, Lorenz System*

- Implemented and compared AM, FM, and chaos modulation for secure signal transmission; applied Bessel-function correction for FM SNR accuracy.
- Demonstrated Lorenz-attractor-based chaos synchronization for encryption/decryption; quantified BER and SNR improvements over conventional AM/FM schemes.

16:1 MUX — VLSI Research | *Cadence Virtuoso, GPDK 180nm, Transmission Gate Logic*

- Designed and simulated a 16:1 MUX using transmission gate (TG) logic in Cadence Virtuoso at GPDK 180nm process.
- Compared TG logic against CMOS static logic on area and power metrics; published findings as VLSI research paper.

Alcohol Intoxication Detection System | *MQ-3 Sensor, LLM, Voice Analysis, Python*

- Fused MQ-3 alcohol sensor readings with LLM-based voice pattern analysis for multi-modal intoxication detection.
- Triggered SMS alerts on positive detection; designed sensor fusion logic to reduce false positives from sensor-only threshold triggers.

Environmental Monitoring System | *ESP32, DHT22, Supabase, Chart.js, Wi-Fi*

- Built a Wi-Fi connected sensor node for continuous temperature, humidity, and air quality monitoring with cloud push to Supabase.
- Developed a dark glassmorphism web dashboard using Chart.js with real-time streaming and historical trend visualization.

Certifications

- **IIT MADRAS SHASTRA 2025-Robo Soccor** | Finalist — 2025
- **NVIDIA DLI — Generative AI** | NVIDIA Deep Learning Institute — 2025
- **ARM-M based Architecture overview** | Microchip University, Microchip — Done

Leadership & Memberships

Team Lead — SIH '25
Closed cabin CO detection and safety protocol system — Top 50 / ∼1,700 SIH Teams — *2026*

Member — IEEE RAS — VIT Chennai
Robotics and Automation Society — *2024 – Present*

Achievements Records

- **Smart India Hackathon (SIH)** — Top 50 of ∼1,700 teams Inter college
- **Akvosphere Embedded Engineer** — Sole embedded engineer designing PCB, firmware, and ML stack for a commercial atmospheric water generation IoT product
- **VLSI Research** — Authored research paper based project on 16:1 MUX design using Transmission Gate logic in Cadence Virtuoso (GPDK 180nm)